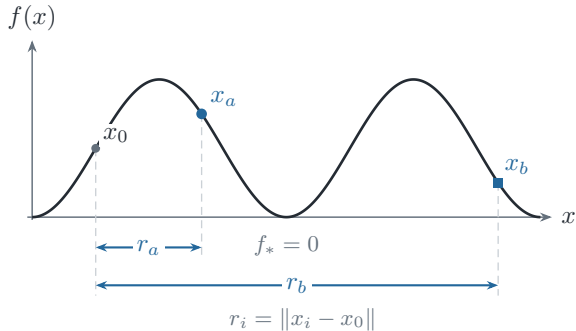


(a) Distance-dependent noise: q -BG

$$\mathbb{E}_{\xi} \|\zeta(x; \xi)\|^p \leq Bq(\|x - x_0\|) + G$$



(b) Gap-dependent noise: q -ABC

$$\mathbb{E}_{\xi} \|\zeta(x; \xi)\|^p \leq Bq(f(x) - f_*) + G$$

